

Sensor Systems and Vehicle Dynamics: Mathematical Foundations and Theory

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Abstract

This document provides the complete mathematical foundation for the sensor simulation and vehicle dynamics model used in autonomous mining vehicle development. We derive the bicycle model kinematics, Ackermann steering geometry, longitudinal dynamics with energy flow, and comprehensive sensor noise models including IMU bias dynamics, GNSS random walk, and coordinate transformations between vehicle body frame, NED frame, and WGS84 geodetic coordinates.

Contents

1 Coordinate Systems and Transformations

1.1 Reference Frames

1.1.1 Global Inertial Frame (World)

Fixed reference frame with origin at simulation start:

- X – East (or initial vehicle forward direction)
- Y – North (or initial vehicle left direction)
- Z – Up (vertical, right-hand rule)

Vehicle state in global frame: (x, y, ψ) where ψ is heading angle (yaw) measured counter-clockwise from X -axis.

1.1.2 Vehicle Body Frame

Moving frame attached to vehicle with origin at rear axle center:

- x_b – Forward (longitudinal)
- y_b – Left (lateral)
- z_b – Up (vertical)

Vehicle velocity in body frame: v (longitudinal speed, m/s).

1.1.3 North-East-Down (NED) Frame

Navigation frame used by GNSS:

- N – North
- E – East
- D – Down (toward Earth center)

1.2 Coordinate Transformations

1.2.1 Body to Global

Position rate of change:

$$\dot{x} = v \cos(\psi) \tag{1}$$

$$\dot{y} = v \sin(\psi) \tag{2}$$

Rotation matrix from body to global:

$$\mathbf{R}_b^g = \begin{bmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{3}$$

1.2.2 Body to NED

Velocity transformation:

$$v_N = v \sin(\psi) \tag{4}$$

$$v_E = v \cos(\psi) \tag{5}$$

$$v_D = 0 \quad (\text{ground vehicle}) \tag{6}$$

1.2.3 Local XY to WGS84

Flat Earth approximation for short distances (< 10 km):

$$\text{meters per degree latitude: } m_\phi = 111,320 \text{ m} \quad (7)$$

$$\text{meters per degree longitude: } m_\lambda = 111,320 \cdot \cos(\phi_0) \text{ m} \quad (8)$$

Position conversion:

$$\phi = \phi_0 + \frac{y}{m_\phi} \quad (9)$$

$$\lambda = \lambda_0 + \frac{x}{m_\lambda} \quad (10)$$

where (ϕ_0, λ_0) is the origin in WGS84 coordinates (latitude, longitude in degrees).

2 Bicycle Model Kinematics

2.1 Model Assumptions

1. **No slip:** Pure rolling condition (wheels do not slide)
2. **Planar motion:** Vehicle remains on flat ground (no pitch/roll)
3. **Point mass:** Neglect rotational inertia about vertical axis
4. **Rear-axle reference:** Kinematic center at rear axle midpoint

2.2 Kinematic Equations

The bicycle model reduces a four-wheeled vehicle to two wheels separated by wheelbase L :

$$\begin{aligned} \dot{x} &= v \cos(\psi) \\ \dot{y} &= v \sin(\psi) \\ \dot{\psi} &= \frac{v}{L} \tan(\delta) \end{aligned} \quad (11)$$

where:

- (x, y) – position of rear axle center in global frame (m)
- ψ – heading angle (rad)
- v – longitudinal velocity (m/s)
- δ – virtual bicycle steering angle (rad)
- L – wheelbase (m)

2.3 Derivation of Yaw Rate

Step 1: Front wheel velocity perpendicular to wheel heading (no-slip):

$$v_f = \frac{v}{\cos(\delta)} \quad (12)$$

Step 2: Lateral component of front wheel velocity:

$$v_{f,\text{lateral}} = v_f \sin(\delta) = v \tan(\delta) \quad (13)$$

Step 3: Angular velocity about rear axle (instantaneous center of rotation):

$$\dot{\psi} = \frac{v_{f,\text{lateral}}}{L} = \frac{v \tan(\delta)}{L} \quad (14)$$

2.4 Turning Radius

From the yaw rate equation:

$$R = \frac{L}{\tan(\delta)} \quad (15)$$

For small angles ($|\delta| < 0.2 \text{ rad} \approx 11^\circ$):

$$\tan(\delta) \approx \delta \implies R \approx \frac{L}{\delta} \quad (16)$$

2.5 Discrete-Time Integration

Using Euler forward integration with timestep Δt :

$$\psi_{k+1} = \psi_k + \Delta t \cdot \frac{v_k}{L} \tan(\delta_k) \quad (17)$$

$$x_{k+1} = x_k + \Delta t \cdot v_k \cos(\psi_k) \quad (18)$$

$$y_{k+1} = y_k + \Delta t \cdot v_k \sin(\psi_k) \quad (19)$$

Stability condition: To prevent integration instability:

$$\Delta t < \frac{L}{v_{\max} \tan(\delta_{\max})} \quad (20)$$

For $L = 2.8 \text{ m}$, $v_{\max} = 60 \text{ m/s}$, $\delta_{\max} = 35^\circ \approx 0.61 \text{ rad}$:

$$\Delta t < 0.067 \text{ s} = 67 \text{ ms} \quad (21)$$

Chosen timestep: $\Delta t = 10 \text{ ms}$ provides $6.7\times$ safety margin.

3 Ackermann Steering Geometry

3.1 Problem Statement

Real vehicles have two front wheels that must turn at different angles to maintain pure rolling (no tire scrubbing). All wheels must rotate about a common **instantaneous center of rotation (ICR)**.

3.2 Geometric Configuration

Parameter	Symbol
Wheelbase	L
Track width (front)	W
Virtual steering angle	δ
Turning radius (rear axle to ICR)	R

3.3 Derivation

Step 1: Turning radius from bicycle model:

$$R = \frac{L}{\tan(\delta)} \quad (22)$$

Step 2: Inner and outer wheel radii:

$$R_{\text{inner}} = R - \frac{W}{2} \quad (23)$$

$$R_{\text{outer}} = R + \frac{W}{2} \quad (24)$$

Step 3: Front wheel angles from geometry:

$$\delta_{\text{inner}} = \arctan\left(\frac{L}{R_{\text{inner}}}\right) = \arctan\left(\frac{L}{R - W/2}\right) \quad (25)$$

$$\delta_{\text{outer}} = \arctan\left(\frac{L}{R_{\text{outer}}}\right) = \arctan\left(\frac{L}{R + W/2}\right) \quad (26)$$

3.4 Turn Direction Convention

For **left turn** ($\delta > 0$):

$$\delta_{FL} = \arctan\left(\frac{L}{R - W/2}\right) \quad (\text{inner, larger angle}) \quad (27)$$

$$\delta_{FR} = \arctan\left(\frac{L}{R + W/2}\right) \quad (\text{outer, smaller angle}) \quad (28)$$

For **right turn** ($\delta < 0$):

$$\delta_{FR} = \arctan\left(\frac{L}{R - W/2}\right) \quad (\text{inner, larger magnitude}) \quad (29)$$

$$\delta_{FL} = \arctan\left(\frac{L}{R + W/2}\right) \quad (\text{outer, smaller magnitude}) \quad (30)$$

3.5 Small Angle Approximation

For $|\delta| \ll 1$:

$$\delta_{\text{inner}} \approx \frac{L}{R - W/2} = \delta \left(1 + \frac{W}{2R}\right) \quad (31)$$

$$\delta_{\text{outer}} \approx \frac{L}{R + W/2} = \delta \left(1 - \frac{W}{2R}\right) \quad (32)$$

4 Longitudinal Dynamics

4.1 Force Balance

Newton's second law in longitudinal direction:

$$m \frac{dv}{dt} = F_{\text{motor}} - F_{\text{brake}} - F_{\text{drag}} - F_{\text{roll}} \quad (33)$$

where m is vehicle mass (kg) and v is longitudinal velocity (m/s).

4.2 Motor Force

Electric motor torque converted to wheel force:

$$F_{\text{motor}} = \frac{T_{\text{motor}} \cdot N_{\text{gear}} \cdot \eta_{\text{drivetrain}}}{r_{\text{wheel}}} \quad (34)$$

where:

- T_{motor} – motor shaft torque (Nm)
- N_{gear} – gear ratio (dimensionless)
- $\eta_{\text{drivetrain}}$ – drivetrain efficiency (typically 0.92)
- r_{wheel} – effective wheel radius (m)

4.3 Power Constraint

Motors are torque-limited at low speed and power-limited at high speed:

$$P_{\text{motor}} = T_{\text{motor}} \cdot \omega_{\text{motor}} \quad (35)$$

Motor angular velocity:

$$\omega_{\text{motor}} = \frac{v}{r_{\text{wheel}}} \cdot N_{\text{gear}} \quad (36)$$

Maximum torque at given speed:

$$T_{\text{motor}}^{\text{max}}(v) = \frac{P_{\text{motor}}^{\text{max}}}{\omega_{\text{motor}}(v)} = \frac{P_{\text{motor}}^{\text{max}} \cdot r_{\text{wheel}}}{v \cdot N_{\text{gear}}} \quad (37)$$

4.4 Aerodynamic Drag

Quadratic drag force:

$$F_{\text{drag}} = \frac{1}{2} \rho C_d A v^2 = C_{\text{drag}} v |v| \quad (38)$$

where:

- ρ – air density (1.225 kg/m³ at sea level)
- C_d – drag coefficient (dimensionless, 0.25–1.2 depending on vehicle)
- A – frontal area (m²)
- $C_{\text{drag}} = \frac{1}{2} \rho C_d A$ – combined drag constant

4.5 Rolling Resistance

Simplified constant friction model:

$$F_{\text{roll}} = C_{\text{roll}} \cdot \text{sign}(v) \quad (39)$$

where C_{roll} is rolling resistance force magnitude (N).

More detailed model:

$$F_{\text{roll}} = \mu_{\text{roll}} \cdot m \cdot g \quad (40)$$

where μ_{roll} is rolling resistance coefficient (0.01–0.03 for asphalt).

4.6 Friction Braking

Brake force proportional to pedal input:

$$F_{\text{brake}} = \frac{p_{\text{brake}}}{100} \cdot \frac{T_{\text{brake}}^{\text{max}}}{r_{\text{wheel}}} \quad (41)$$

where $p_{\text{brake}} \in [0, 100]$ is brake pedal percentage.

4.7 Velocity Update

Acceleration:

$$a = \frac{F_{\text{motor}} - F_{\text{brake}} - F_{\text{drag}} - F_{\text{roll}}}{m} \quad (42)$$

Discrete velocity update:

$$v_{k+1} = v_k + \Delta t \cdot a_k \quad (43)$$

Zero-crossing logic: Prevent oscillation near standstill:

$$\text{if } \text{sign}(v_{k+1}) \neq \text{sign}(v_k) \text{ and } |v_{k+1}| < \epsilon_v, \text{ then } v_{k+1} = 0 \quad (44)$$

where $\epsilon_v = 0.3$ m/s is velocity threshold.

5 Lateral Dynamics and Friction Limits

5.1 Centripetal Acceleration

During cornering, lateral acceleration is:

$$a_{\text{lat}} = v \cdot \dot{\psi} = \frac{v^2}{L} \tan(\delta) \quad (45)$$

5.2 Friction Circle Constraint

Maximum lateral acceleration limited by tire-road friction:

$$|a_{\text{lat}}| \leq \mu g \quad (46)$$

where:

- μ – lateral friction coefficient (0.1–0.9 depending on surface)
- $g = 9.81$ m/s² – gravitational acceleration

5.3 Understeer Implementation

When friction limit exceeded, scale yaw rate:

$$\dot{\psi}_{\text{actual}} = \dot{\psi}_{\text{commanded}} \cdot \min \left(1, \frac{\mu g}{|a_{\text{lat}}|} \right) \quad (47)$$

6 Battery and Energy Model

6.1 State of Charge Dynamics

Battery state of charge (SOC) evolution:

$$\frac{d(\text{SOC})}{dt} = -\frac{I(t)}{Q_{\text{total}}} \quad (48)$$

where:

- $\text{SOC} \in [0, 1]$ – normalized state of charge
- I – battery current (A), positive = discharge
- Q_{total} – total battery capacity (Ah)

In terms of power:

$$\frac{d(\text{SOC})}{dt} = -\frac{P(t)}{C_{\text{kWh}} \cdot 1000 \text{ W}} \quad (49)$$

where C_{kWh} is capacity in kWh and P is power in W.

6.2 Power Flow Directions

Discharge (acceleration):

$$P_{\text{battery}} = \frac{P_{\text{motor}}}{\eta_{\text{discharge}}} \quad (50)$$

$$I_{\text{battery}} = \frac{P_{\text{battery}}}{V_{\text{battery}}} \quad (\text{positive}) \quad (51)$$

Charge (regenerative braking):

$$P_{\text{battery}} = P_{\text{motor}} \cdot \eta_{\text{charge}} \quad (52)$$

$$I_{\text{battery}} = \frac{P_{\text{battery}}}{V_{\text{battery}}} \quad (\text{negative}) \quad (53)$$

6.3 Discrete SOC Update

Energy flow in timestep Δt :

$$\Delta E = P \cdot \Delta t \quad (\text{Watt-seconds} = \text{Joules}) \quad (54)$$

Convert to Watt-hours:

$$\Delta E_{\text{Wh}} = \frac{P \cdot \Delta t}{3600} \quad (55)$$

SOC change:

$$\Delta(\text{SOC}) = -\frac{\Delta E_{\text{Wh}}}{C_{\text{kWh}} \cdot 1000} \quad (56)$$

Update with saturation:

$$\text{SOC}_{k+1} = \text{clamp}(\text{SOC}_k + \Delta(\text{SOC}), \text{SOC}_{\min}, \text{SOC}_{\max}) \quad (57)$$

7 Sensor Noise Models

7.1 Gaussian White Noise

Additive zero-mean Gaussian noise:

$$y = x_{\text{truth}} + \mathcal{N}(0, \sigma^2) \quad (58)$$

where $\mathcal{N}(0, \sigma^2)$ is normal distribution with standard deviation σ .

Implementation: Box-Muller transform:

$$U_1, U_2 \sim \mathcal{U}(0, 1) \quad (\text{uniform random}) \quad (59)$$

$$Z_1 = \sqrt{-2 \ln U_1} \cos(2\pi U_2) \quad (60)$$

$$Z_2 = \sqrt{-2 \ln U_1} \sin(2\pi U_2) \quad (61)$$

where $Z_1, Z_2 \sim \mathcal{N}(0, 1)$.

Scaled noise: $\sigma Z_1 \sim \mathcal{N}(0, \sigma^2)$.

7.2 Random Walk

Brownian motion (Wiener process):

$$dx = \sigma_{\text{rw}} dW \quad (62)$$

where $dW \sim \mathcal{N}(0, dt)$ is Wiener increment.

Discrete implementation:

$$x_{k+1} = x_k + \sigma_{\text{rw}} \sqrt{\Delta t} \cdot \mathcal{N}(0, 1) \quad (63)$$

Example: GNSS position drift with $\sigma_{\text{rw}} = 0.1 \text{ m}/\sqrt{\text{s}}$.

7.3 Gauss-Markov Process

First-order Markov process (exponentially correlated noise):

$$\frac{db}{dt} = -\frac{1}{\tau}b + w(t) \quad (64)$$

where:

- b – bias state
- τ – correlation time constant (s)
- $w(t) \sim \mathcal{N}(0, \sigma_w^2)$ – white noise driving term

Discrete solution:

$$b_{k+1} = e^{-\Delta t/\tau} b_k + \sigma_b \sqrt{1 - e^{-2\Delta t/\tau}} \cdot \mathcal{N}(0, 1) \quad (65)$$

where σ_b is steady-state bias standard deviation.

Steady-state variance:

$$\sigma_b^2 = \frac{\tau \sigma_w^2}{2} \quad (66)$$

8 IMU Sensor Model

8.1 Gyroscope Model

Measured angular velocity:

$$\omega_{\text{meas}} = \omega_{\text{truth}} + b_{\text{gyro}} + n_{\text{gyro}} \quad (67)$$

where:

- ω_{truth} – true angular velocity (rad/s)
- b_{gyro} – time-varying bias (Gauss-Markov, $\tau = 1800$ s)
- $n_{\text{gyro}} \sim \mathcal{N}(0, \sigma_{\text{gyro}}^2)$ – white noise ($\sigma = 0.1/\text{s}$)

Bias evolution:

$$b_{\text{gyro},k+1} = \beta b_{\text{gyro},k} + \sigma_b \sqrt{1 - \beta^2} \cdot \mathcal{N}(0, 1) \quad (68)$$

where $\beta = \exp(-\Delta t/\tau)$.

8.2 Accelerometer Model

Measured acceleration (body frame):

$$\mathbf{a}_{\text{meas}} = \mathbf{a}_{\text{truth}} + \mathbf{b}_{\text{accel}} + \mathbf{n}_{\text{accel}} \quad (69)$$

Truth values in body frame:

$$a_{x,\text{truth}} = \frac{dv}{dt} \quad (\text{longitudinal acceleration}) \quad (70)$$

$$a_{y,\text{truth}} = v \cdot \dot{\psi} \quad (\text{centripetal acceleration}) \quad (71)$$

$$a_{z,\text{truth}} = -g = -9.81 \text{ m/s}^2 \quad (\text{gravity, body Z up}) \quad (72)$$

Bias: Gauss-Markov with $\tau = 3600$ s (1 hour), $\sigma_b = 0.02 \text{ m/s}^2$.

White noise: $\sigma_{\text{accel}} = 0.05 \text{ m/s}^2$.

8.3 Gravity Compensation

Accelerometers measure **specific force** (excluding gravity). For a stationary vehicle:

$$\mathbf{a}_{\text{meas}} = -\mathbf{g}_{\text{body}} + \text{bias} + \text{noise} \quad (73)$$

where $\mathbf{g}_{\text{body}} = [0, 0, -9.81]^\top$ m/s² in body frame (Z-up).

During motion:

$$\mathbf{a}_{\text{meas}} = \mathbf{a}_{\text{vehicle}} - \mathbf{g}_{\text{body}} + \text{bias} + \text{noise} \quad (74)$$

9 GNSS Sensor Model

9.1 Position Measurement

Latitude and longitude with random walk drift:

$$\phi_{\text{meas}} = \phi_{\text{truth}} + d_\phi + \mathcal{N}(0, \sigma_\phi^2) \quad (75)$$

$$\lambda_{\text{meas}} = \lambda_{\text{truth}} + d_\lambda + \mathcal{N}(0, \sigma_\lambda^2) \quad (76)$$

where:

- d_ϕ, d_λ – random walk drift ($\sigma_{\text{rw}} = 0.1$ m/ $\sqrt{\text{s}}$)
- $\sigma_\phi = 2.0$ m / m_ϕ – white noise (latitude)
- $\sigma_\lambda = 2.0$ m / m_λ – white noise (longitude)

9.2 Velocity Measurement (NED Frame)

North-East velocity:

$$v_{N,\text{meas}} = v_N + \mathcal{N}(0, 0.1^2) \quad (77)$$

$$v_{E,\text{meas}} = v_E + \mathcal{N}(0, 0.1^2) \quad (78)$$

where $v_N = v \sin(\psi)$ and $v_E = v \cos(\psi)$ from body velocity transformation.

9.3 Circular Error Probable (CEP)

The 50% probability error circle radius:

$$\text{CEP} = 0.59(\sigma_N + \sigma_E) \approx 1.2 \text{ m} \quad (79)$$

for $\sigma_N = \sigma_E = 2.0$ m.

10 Wheel Speed Encoder Model

10.1 Quantization Model

Encoder with N_{ticks} pulses per revolution:

$$\text{Resolution} = \frac{2\pi}{N_{\text{ticks}}} \text{ rad} \quad (80)$$

Quantized measurement:

$$\omega_{\text{quant}} = \left\lfloor \frac{\omega_{\text{truth}}}{\text{Resolution}} \right\rfloor \cdot \text{Resolution} \quad (81)$$

where $\lfloor \cdot \rfloor$ denotes rounding to nearest integer.

10.2 Total Encoder Noise

Combined quantization and sensor noise:

$$\omega_{\text{meas}} = \omega_{\text{quant}} + \mathcal{N}(0, \sigma_{\omega}^2) \quad (82)$$

where $\sigma_{\omega} = 0.5$ rad/s (white noise).

10.3 Per-Wheel Variation

Simulate tire pressure and slip variations:

$$\omega_i = \omega_{\text{meas}} \cdot (1 + u_i) \quad (83)$$

where $u_i \sim \mathcal{U}(-0.02, +0.02)$ is uniform random variable (2% variation).

11 Radar Sensor Model

11.1 Range Measurement

Simulated stationary target at fixed distance:

$$r_{\text{meas}} = r_{\text{truth}} + \mathcal{N}(0, \sigma_r^2) \quad (84)$$

where $r_{\text{truth}} = 50.0$ m and $\sigma_r = 0.2$ m.

11.2 Doppler Velocity

Relative radial velocity (closing speed):

$$v_{\text{rel}} = v_{\text{ego}} \cos(\theta) \quad (85)$$

where θ is angle between vehicle heading and line-of-sight to target.
For target dead ahead ($\theta = 0$):

$$v_{\text{meas}} = v_{\text{truth}} + \mathcal{N}(0, \sigma_v^2) \quad (86)$$

where $\sigma_v = 0.1$ m/s.

11.3 Angle Measurement

Azimuth angle to target:

$$\alpha_{\text{meas}} = \alpha_{\text{truth}} + \mathcal{N}(0, \sigma_{\alpha}^2) \quad (87)$$

where $\alpha_{\text{truth}} = 0$ (dead ahead) and $\sigma_{\alpha} = 0.5$.

12 Validation and Verification

12.1 Statistical Tests

12.1.1 Chi-Squared Goodness of Fit

Test if noise distribution matches Gaussian:

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \quad (88)$$

where O_i are observed counts and E_i are expected counts in histogram bins.

12.1.2 Allan Variance

Characterize gyroscope noise:

$$\sigma_A^2(\tau) = \frac{1}{2(N-1)} \sum_{i=1}^{N-1} (\bar{\omega}_{i+1}(\tau) - \bar{\omega}_i(\tau))^2 \quad (89)$$

where $\bar{\omega}_i(\tau)$ is average over interval τ .

12.2 Coordinate Transform Verification

Round-trip test: Body $\rightarrow$ NED $\rightarrow$ Body should preserve magnitude:

$$v = \sqrt{v_N^2 + v_E^2} \quad (\text{Pythagorean theorem}) \quad (90)$$

Gravity check: Stationary vehicle should measure:

$$\mathbf{a}_{\text{meas}} = [0, 0, -9.81]^\top + \text{small noise/bias} \quad (91)$$

13 Conclusion

This mathematical framework provides the foundation for:

- High-fidelity vehicle dynamics simulation
- Realistic sensor noise modeling for ML/AI training
- Coordinate transformations between body, NED, and WGS84 frames
- Extended Kalman Filter (EKF) development and tuning
- Sensor fusion algorithm validation

All models are validated against industry standards and real sensor specifications.

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